BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type (Group-I)

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Inter, Part I, 2022 Lahore Board. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Pasteurization technique is widely used for preservation of:

(A) Water
(B) Meal
(C) Milk and milk products
(D) Vaccines

Q2. The compounds which on hydrolysis yield polyhydroxy aldehyde or ketone with units are:

(A) Carbohydrates
(B) Proteins
(C) Lipids
(D) Nucleic acids

Q3. The raw material from which coenzymes are made:

(A) Proteins
(B) Nucleic acids
(C) Vitamins
(D) Carbohydrates

Q4. Organelle involved in the synthesis of ATP is:

(A) Ribosomes
(B) Nucleus
(C) Centriole
(D) Mitochondria

Q5. Prions are made up of:

(A) Nucleic acids
(B) Proteins
(C) Lipids
(D) Carbohydrates

Q6. A cube of eight cocci is called:

(A) Diplococci
(B) Streptococci
(C) Tetrad
(D) Sarcina

Q7. Tests of foraminifera are made of:

(A) Potassium
(B) Calcium
(C) Silica
(D) Iron

Q8. Corn smut is caused by:

(A) Candida albicans
(B) aspergillus fumigatus
(C) Penicillium notatum
(D) Ustilago maydis

Q9. The female gametophyte of flowering plant consists of ____ cells:

(A) 2
(B) 4
(C) 7
(D) 8

Q10. Blastopore forms anus in:

(A) Echinodermata
(B) Annelida
(C) Nematoda
(D) Mollusca

Q11. The parasite which produces anticoagulant to prevent blood clotting is:

(A) Hook worm
(B) Pin worm
(C) Ascaris
(D) Fasciola

Q12. Number of NADH produced by passing one pyruvate molecule through Kreb's Cycle pyruvic acid oxidation is / are:

(A) 1
(B) 2
(C) 3
(D) 4

Q13. Which part of light spectrum produces more oxygen during photosynthesis:

(A) Blue
(B) Green
(C) Yellow
(D) Red

Q14. Enterokinase enzymes is secreted by lining of:

(A) Pancreas
(B) Liver
(C) Duodenum
(D) Stomach

Q15. Smoker's cough cause:

(A) Asthma (B) Emphysema

(C) Cancer (D) Tuberculosis

Q16. The weight of blood in a man of 60 kg is:

(A) 5 kg (B) 10 kg

(C) 15 kg (D) 20 kg

Q17. The cells which supply ATP and proteins to sieve tubes are:

(A) Fibers (B) Companion cells

(C) Scleriedes (D) Guard cells

Essay Type (Group-I)

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

QSECTION – I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Why lipids store double amount of energy than carbohydrates?

ii Why inhibitors affect enzyme function? Mention with examples.

iii Why binding site and catalytic site are important for enzymes?

iv Why enzymes are affected by extreme changes in pH?

v Enlist six plant diseases caused by fungi.

vi Define nuclear mitosis in fungi.

vii Give four diagnostic characters of mammals.

viii What do you know about harmful insects?

ix Differentiate between polyp and medusa.

x Define radial and bilateral symmetry with examples.

xi What is compensation point? When it occurs?

xii How action spectrum can be obtained?

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i Define population and give its four attributes.

ii What do you know about integrated disease management?

iii Why plasma membrane do not allow all the substances to cross it?

iv Which organelle of the cell engulfs the foreign objects, also give the purpose of this process?

v Why protists are considered as polyphyletic? Give two examples of animal like protists.

vi How foraminiferous have poles in their shells? By which way shell is transformed into chalk?

vii What do you know about amoeba?

viii How red algae are differentiated from green algae?

ix What is phylogenetic system of classification?

x Give characteristics of bryophytes (briefly).

xi Write factors which are responsible for bleeding in plants.

xii Define immunity and name its types.

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

i On the basis of morphology, how viruses are classified?

ii What are mesosomes? Describe their functions.

iii Write name of four parts of digestive system of cockroach.

iv How our oral cavity selects food for further digestion?

v Differentiate between peristalsis and antiperistalsis.

vi What is rubisco? Give its function.

vii Enlist types of respiration in frog.

viii Differentiate between diaphragm and pleura.

ix What is myoglobin? How it differs from haemoglobin?

QSECTION-II. Note: Attempt any THREE questions.

QQ.5(a). (Marks 4) The environment of Pakistan is deteriorating day by day, suggest various measures to conserve it.

QQ.5(b). (Marks 4) Write down any eight functions of blood.

QQ.6(a). (Marks 4) What are polysaccharides? Give details of some biologically important polysaccharides.

QQ.6(b). Describe economic gains due to fungi.

QQ.7(a). How bacteria are classified on the basis of nutrients and energy trapping methods?

QQ.7(b). How plants applied their different features to live successfully on land?

QQ.8(a). What is nomenclature, describe its importance with the help of examples.

QQ.8(b). Draw the outlines of Kreb's Cycle.

QQ.9(a). What do you know about endoplasmic reticulum? Explain with diagram.

QQ.9(b). Food poisoning and obesity are diseases related to nutrition. Discuss.

GROUP – II

Objective Type (Group-II)

Time Allowed: 20 Min. | **Maximum Marks:** 17

> Note: Inter, Part I, 2022 Lahore Board. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. A plant requires nitrogen and sulphur for its:

(A) DNA replication
(B) cell wall
(C) Enzymes
(D) Starch

Q2. Histoplasmosis is a disease of:

(A) Liver
(B) Skin
(C) Lung
(D) Brain

Q3. Prions are made up of:

(A) Lipid
(B) DNA
(C) Protein
(D) RNA

Q4. Which one of the following does not belong to phylum platyhelminthes:

(A) Crab
(B) Planaria
(C) Liver fluke
(D) Tape worm

Q5. Clitoria termatea is used against:

(A) Dog bite
(B) Scorpion bite
(C) Snake bite
(D) Wasp bite

Q6. Fats and oils have specific gravity of about:

(A) 0.8
(B) 8
(C) 0.08
(D) 0.008

Q7. Following are organic nutrients present in the blood except:

(A) Glucose
(B) Fats
(C) Amino acids
(D) Sodium chloride

Q8. Choose the smallest bacteria:

(A) E.Coli
(B) Mycoplasma
(C) Spinochctes
(D) Staphylocci

Q9. Chlorophyll molecule contains following except:

(A) Magnesium
(B) Iron
(C) Porphyrin ring
(D) phytol

Q10. The study of distribution of animals in nature is called:

(A) Wildlife
(B) Zoogeography
(C) Geography
(D) Biodiversity

Q11. The largest invertebrate is:

(A) Earthworm
(B) Giant squid
(C) Ascaris
(D) Star fish

Q12. Co-enzyme is closed related to:

(A) Vitamins
(B) Water
(C) Minerals
(D) Lipids

Q13. How much air can hold fully inflated lungs:

(A) 4 litres
(B) 4.5 litres
(C) 5 litres
(D) 5.5 litres

Q14. How many CO2 molecules are produced from complete breakdown of one pyruvate:

(A) 2
(B) 4
(C) 6
(D) 3

Q15. Closed circulatory system is present in animals except:

(A) Squid
(B) Spiders
(C) Octopus
(D) Fish

Q16. Example of apicomplexans is:

(A) Amoeba
(B) Paramecium
(C) Bacteria
(D) Plasmodium

Q17. The number of nuclear pores in undifferentiated cells (such as eggs) are:

(A) 3000 (B) 40,000

(C) 5000 (D) 30,000

Essay Type (Group-II)

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

QSECTION – I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i While role is played by cellulose digesting enzymes in plant eating animals, discuss briefly.

ii At constant temperature and pH, how rate of reaction can be doubled?

iii In which way inhibitors stop catalytic activity of the enzymes? Give one example.

iv Why enzymes remain unaltered after the formation of products?

v How rust is differentiated from smut?

vi What is parasexuality in fungi?

vii Name the phylum of these animals, octopus, tape worm, leech and dolphin.

viii Give two differences between protostomes and deuterostomes.

ix What do you know about pseudocoelomates?

x Define polymorphism, also give an example.

xi Differentiate between photosystem-I and photosystem-II.

xii Write name of four stages of cellular respiration.

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i Compare radiotherapy and gene therapy to control diseases.

ii Differentiate between embryonic and organism cloning.

iii How cell membrane is differentially permeable membrane? Justify it.

iv What is the factory of synthesis of protein and ribosomes?

v Basically kingdom protista is defined by exclusion, how?

vi How slime moulds adopt unfavourable and favourable conditions?

vii Why phytophthora infestans is infamous in human history?

viii What type of pigments are found in rhodophyta and chlorophyta?

ix Define double fertilization. Give its importance.

x Write scientific names of potato and amaltas.

xi Describe apoplast pathway.

xii What are granulocytes?

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

i Write down contributions of E-Chatton.

ii Shortly write down the structure of bacteria.

iii Write down the role of nitrogen and phosphorus in plant growth.

iv Write few lines on filter feeders.

v Differentiate cardiac and pyloric sphincter with reference to their function.

vi Write down CO2 concentration in arterial and venous blood.

vii What is the role of partial pressure of O2 during shock?

viii Write short note on asthma.

ix How rubisco is converted into serine?

QSECTION-II. Note: Attempt any THREE questions.

QQ.5(a). (Marks 4) Give the details of biological conservation and protection of environment.

QQ.5(b). (Marks 4) What is immunity? Discuss its major types.

QQ.6(a). (Marks 4) What is RNA? Describe its different types.

QQ.6(b). (Marks 4) Discuss the economic gains due to fungi.

QQ.7(a). (Marks 4) Cell membrane and nuclear material of bacteria differ from that of eukaryotic.

QQ.7(b). (Marks 4) How the cell life of an angiospermic plant differs from a gymnospermic plant?

QQ.8(a). (Marks 4) Illustrate the life cycle of bacteriophage diagrammatically.

QQ.8(b). (Marks 4) Give in detail the carbon fixation and reduction phase of Calvin Cycle.

QQ.9(a). (Marks 4) Discuss structure and chemical composition of cell wall.

QQ.9(b). (Marks 4) Explain the process of digestion in oral cavity of man.